Cross-Encoder Re-Ranking for High Precision Retrieval

Bi-Encoders vs Cross-Encoders

Bi-encoders encode queries and documents independently for fast ANN search. Cross-encoders process query and document pairs simultaneously through full self-attention layers, capturing complex token-level interactions.

Two-Stage Retrieval Pipeline

First-stage candidate retrieval fetches top-100 candidates via fast vector/BM25 search. Second-stage cross-encoders re-rank candidate pairs to select top-5 high-precision context chunks for LLM inference.

Latency and Throughput Trade-offs

Cross-encoders incur O(N) inference cost per query. Batching and GPU acceleration keep re-ranking latency under 30ms for 50 candidate documents.